

What the boundary algebra of p -adic AdS/CFT does not see

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Abstract

In p -adic AdS/CFT the bulk is the Bruhat–Tits tree of PGL_2 over a local field and the boundary is $\mathbb{P}^1(F)$. The boundary carries two standard operator algebras, the crossed products $C(\mathbb{P}^1(F)) \rtimes \Gamma$ and $L^\infty(\mathbb{P}^1(F)) \rtimes \Gamma$, and both have been completely determined: the first is a Cuntz–Krieger algebra with $K_0 = \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$ depending only on the rank of Γ , and the second is the hyperfinite factor of type III_λ with $\lambda = q^{-2}$ or q^{-1} according as the quotient graph is bipartite or not (Robertson, 2005). Neither result has ever been cited in the holography literature, and no p -adic AdS/CFT paper has used an operator-algebraic invariant at all. We import them, and add one arithmetic layer. The bulk Hecke–Laplacian channel has $K_0 = \mathbb{Z} \oplus \mathrm{Jac}(Y)$, the critical group of the quotient graph, whose order is a product of Frobenius traces $\frac{1}{n} \prod_f (p+1 - a_p(f))$. We exhibit two cubic graphs on eight vertices whose boundary algebras agree in K -theory *and* in von Neumann type, and whose bulk critical groups have coprime orders $256 = 2^8$ and $363 = 3 \cdot 11^2$: the boundary data do not determine the bulk arithmetic, not even which primes divide it. This is strictly stronger than Robertson’s own counterexample, which the type does separate. We do not conclude that holography fails: Cornelissen and Marcolli proved that the boundary quantum statistical mechanical system, with its Busemann time evolution, reconstructs the graph completely, so what we exhibit is a precise gap in a hierarchy whose top is a reconstruction theorem. Finally we observe that $\lambda \neq 1$ is itself a diagnostic. Continuum holography is forced to type III_1 by theorems (mixing; half-sided modular inclusion); p -adic holography is type III_λ with $\lambda < 1$, so its modular flow is periodic and its boundary algebra is not a Haag–Kastler algebra. All computations are exact and machine-verified.

1 Introduction

p -adic AdS/CFT [16, 18] replaces anti-de Sitter space by the Bruhat–Tits tree X of $\mathrm{PGL}_2(F)$, F a non-archimedean local field with residue field of size q , and the conformal boundary by $\mathbb{P}^1(F)$, the space of ends of X . A discrete torsion-free cocompact lattice $\Gamma \leq \mathrm{PGL}_2(F)$ — free of finite rank r — plays the role of a quotient identification, exactly as a Schottky group does for BTZ [19], and $Y = \Gamma \backslash X$ is a finite $(q+1)$ -regular graph on n vertices. The field has matured on correlators, tensor networks, Wilson lines and finite temperature [17, 24, 19, 25], but it has remained kinematic in one specific respect: no invariant from operator algebras has ever been computed in it.

That is unfortunate, because the relevant invariants were computed twenty years ago, in a literature the holography community does not read. Robertson [27] determined both boundary

algebras of exactly this situation, and Cornelissen and Marcolli [9] determined precisely how much of the bulk each one remembers. The purpose of this note is to import those results, to add one layer they do not contain, and to draw one physical consequence.

What is imported. For Γ a free uniform tree lattice of rank r , Robertson proved [27, Thm. 1] that $C(\partial X) \rtimes \Gamma$ is a Cuntz–Krieger algebra with

$$K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)\mathbb{Z}, \quad K_1 \cong \mathbb{Z}^r, \quad (1)$$

[1] generating the torsion summand — so by Kirchberg–Phillips the algebra *depends only on r* . (The same K -groups were obtained independently in [8].) He proved further [27, Thm. 2, Cor. 1] that

$$L^\infty(\partial X) \rtimes \Gamma \cong \text{hyperfinite III}_\lambda, \quad \lambda = \begin{cases} q^{-2} & \Gamma \subset \text{PSL}_2(F) \text{ (} Y \text{ bipartite)}, \\ q^{-1} & \text{otherwise,} \end{cases} \quad (2)$$

the ratio set being generated by the Busemann cocycle; the rank-one case goes back to Ramagge and Robertson [26] and the general hyperbolic-group statement to [20, 3]. Cornelissen and Marcolli [9] then showed that the C^* -level invariants “reconstruct only topological information” — nine natural equivalences all reduce to equality of the first Betti number — while the boundary *quantum statistical mechanical* system, $C(\partial X) \rtimes \Gamma$ equipped with the Busemann time evolution, reconstructs Y up to isomorphism.

What is added. The bulk of this geometry has an invariant that none of these papers considers. The Hecke–Laplacian $\Delta_p = (q+1)I - T_p$ of Y has

$$\text{coker}_{\mathbb{Z}} \Delta_p \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad |\text{Jac}(Y)| = \frac{1}{n} \prod_f (p+1 - a_p(f)), \quad (3)$$

where $\text{Jac}(Y)$ is the critical (sandpile) group [2, 1] and the product runs over the weight-two newforms attached to Γ , so that $|\text{Jac}(Y)|$ is a product of Frobenius traces [7, I]. The critical group has appeared in physics — as the global symmetry group of fracton models on graphs [14, 15] — but never in holography. Our contribution is Theorem 3: the boundary data of (1) and (2) *together* fail to determine (3), and fail badly enough that two bulks in the same boundary class can have coprime arithmetic.

What we do not claim. Not that holography fails. [9] is a reconstruction theorem: keep the time evolution and the bulk comes back. What Theorem 3 locates is a gap between two rungs of a ladder whose top rung is a positive result. Nor do we claim (1) or (2); both are [27].

2 Setting

$Y = \Gamma \backslash X$ is a finite connected $(q+1)$ -regular graph, $n = |V_Y|$, $m = (q+1)n/2$, and

$$r = \text{rank } \Gamma = m - n + 1, \quad r - 1 = m - n = \frac{(q-1)n}{2}.$$

The *boundary channel* is the crossed product by the Γ -action on ∂X ; concretely $C(\partial X) \rtimes \Gamma \cong \mathcal{O}_{A^{\text{nb}}}$ with A^{nb} the non-backtracking matrix on the $2m$ oriented edges of Y [27], and $K_0 = \text{coker}(I - (A^{\text{nb}})^t)$, $K_1 = \ker(I - (A^{\text{nb}})^t)$ by [10]. The *bulk channel* is Δ_p on the n vertices, with (3).

Remark 1 (Two relations, two types). Equation (2) concerns the orbit relation of Γ on ∂X . It must be distinguished from the tail equivalence relation on the Cantor limit $\varprojlim_k V_{Y_k}$ of a tower of covers, whose ratio set is all of $q^{\mathbb{Z}} \cup \{0\}$ and which therefore gives type $\text{III}_{1/q}$ unconditionally [7, III]. The orbit relation of a type-preserving lattice realises only $q^{2\mathbb{Z}}$, because the Busemann cocycle of such an element has even value; this is the source of the dichotomy in (2) and it is easy to conflate the two.

Proposition 2. *The boundary K -theory (1) is a function of (q, n) alone, and the von Neumann type (2) is a function of q and the bipartiteness of Y alone. In particular all $(q+1)$ -regular graphs on n vertices with the same bipartiteness have the same boundary data.*

Proof. $r - 1 = (q - 1)n/2$ and $r = (q - 1)n/2 + 1$ depend only on (q, n) ; (2) is as stated. $\square$

3 The bulk is not determined

Theorem 3. *There are two cubic graphs Y_A, Y_B on eight vertices with*

- *the same boundary K -theory, $K_0 = \mathbb{Z}^5 \oplus \mathbb{Z}/4$ and $K_1 = \mathbb{Z}^5$;*
- *the same von Neumann type, both being non-bipartite and hence $\text{III}_{1/q}$;*
- *coprime bulk critical groups,*

$$\text{Jac}(Y_A) = \mathbb{Z}/4 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16, \quad |\text{Jac}(Y_A)| = 256 = 2^8,$$

$$\text{Jac}(Y_B) = \mathbb{Z}/33 \oplus \mathbb{Z}/11, \quad |\text{Jac}(Y_B)| = 363 = 3 \cdot 11^2.$$

Explicitly, with vertices $0, \dots, 7$,

$$E(Y_A) = \{03, 06, 07, 12, 15, 16, 24, 25, 34, 37, 45, 67\},$$

$$E(Y_B) = \{02, 03, 07, 14, 16, 17, 24, 25, 34, 35, 56, 67\}.$$

Consequently the boundary K -theory and the boundary von Neumann type, taken together, do not determine $|\text{Jac}(Y)|$, nor the set of primes dividing it, nor — by (3) — the Frobenius traces $a_p(f)$ of the associated automorphic forms.

Proof. Direct computation of the three invariants in exact integer arithmetic; this is check (P) of Section 5. Both graphs contain an odd cycle, hence are non-bipartite. $\square$

Remark 4 (Comparison with [27]). Robertson gives his own counterexample to boundary rigidity [27, Ex. 1.5]: the theta graph and the dumbbell graph, both with $\pi_1 = F_2$, non-isomorphic, with the same $C(\partial X) \rtimes \Gamma$, realised concretely inside $\text{PGL}_2(\mathbb{Q}_2)$. His example is separated by the von Neumann type, $\text{III}_{1/4}$ against $\text{III}_{1/2}$ — indeed that is his point, that the measure-theoretic invariant is finer. Theorem 3 defeats both invariants at once, and does so in a way that is arithmetically meaningful rather than merely graph-theoretic.

Table 1 shows the two channels on a list of standard quotients, and Table 2 shows how much bulk arithmetic a single boundary class holds.

Y	q	n	r	$K_0(\text{boundary})$	vN type	$ \text{Jac}(Y) $
K_4	2	4	3	$\mathbb{Z}^3 \oplus \mathbb{Z}/2$	$\text{III}_{1/q}$	16
$K_{3,3}$	2	6	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/3$	III_{1/q^2}	81
prism $C_3 \times K_2$	2	6	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/3$	$\text{III}_{1/q}$	75
prism $C_4 \times K_2$	2	8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$	III_{1/q^2}	384
Möbius M_8	2	8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$	$\text{III}_{1/q}$	392
Möbius M_{10}	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	III_{1/q^2}	1815
prism $C_5 \times K_2$	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	1805
Petersen	2	10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	2000
Heawood	2	14	8	$\mathbb{Z}^8 \oplus \mathbb{Z}/7$	III_{1/q^2}	50421
K_5	3	5	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$	$\text{III}_{1/q}$	125
K_6	4	6	10	$\mathbb{Z}^{10} \oplus \mathbb{Z}/9$	$\text{III}_{1/q}$	1296

Table 1: The two channels. $K_1 = \mathbb{Z}^r$ throughout. Within a block of equal (q, n) the boundary K -theory is constant; the type refines it by bipartiteness; $|\text{Jac}(Y)|$ is constrained by neither. Compare prism $C_5 \times K_2$ and Petersen: same (q, n) , same type, $|\text{Jac}| = 1805$ against 2000. All entries exact.

n	r	$K_0(\text{boundary})$	distinct	$ \text{Jac} $	found	range
8	5	$\mathbb{Z}^5 \oplus \mathbb{Z}/4$			5	256 – 392
10	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/5$			16	960 – 2000
12	7	$\mathbb{Z}^7 \oplus \mathbb{Z}/6$			38	2808 – 9747
14	8	$\mathbb{Z}^8 \oplus \mathbb{Z}/7$			54	12100 – 46998

Table 2: Bulk critical groups inside a single boundary K -theory class, from 60 random cubic quotients at each n . Sampled, not exhaustive: the true counts are at least these.

4 The type as a physical diagnostic

The recent algebraic program in holography [22, 23, 29, 5, 6] turns on the fact that the large- N algebra of a holographic CFT, and the algebra of a black hole exterior, are of type III_1 , and that crossed products by the modular flow convert them to semifinite type II, which is what makes generalized entropy finite. Type III_1 is not an accident there: it is forced. If the correlation functions decay — if the algebra is generated by mixing operators in a KMS state — the algebra must be III_1 [12]; and the half-sided modular inclusion structure that [22, 23] use forces III_1 by Wiesbrock’s theorem [28]. Local algebras of relativistic quantum field theory are the hyperfinite III_1 factor [4].

Equation (2) says that p -adic holography is not in that class. Its boundary algebra is III_λ with $\lambda = q^{-1}$ or q^{-2} , a Powers factor. The physical content of $\lambda < 1$ is that the modular flow is *periodic*: the flow of weights is the translation flow on $\mathbb{R}/T\mathbb{Z}$ with

$$T = \frac{2\pi}{\log \lambda^{-1}} = \frac{2\pi}{\kappa \log q}, \quad \kappa \in \{1, 2\},$$

so that the modular Hamiltonian has spectrum in $(\log q^\kappa)\mathbb{Z}$ rather than a continuum. A III_λ algebra has a nontrivial discrete invariant — Connes’ S invariant is $\lambda^\mathbb{Z} \cup \{0\}$ — where a III_1 algebra has none.

We draw one conclusion and refrain from a second. The conclusion: the type is a sharp, computable obstruction to any claim that a p -adic holographic boundary theory is a local

quantum field theory in the Haag–Kastler sense, and it identifies precisely what would have to change — the residue field would have to be made continuous, $q \rightarrow 1$ — for the III_1 of [22, 29] to be recovered. What we refrain from: we do not propose a p -adic analogue of the crossed-product construction of [29]. For III_λ with $\lambda < 1$ the crossed product by the modular flow is a type II_∞ algebra as well, but the relevant flow is periodic, and whether the resulting trace has anything to do with an entropy is a question we have not investigated.

5 Verification

The script `boundary_blind.py` accompanying this note runs the checks of Table 3 in 0.3s, all passing; its output is archived as `boundary_blind_output.txt`. Arithmetic is exact integer throughout except the two rows marked *sampled*.

key	check	mode
(K)	(1) holds on 11 standard quotients, including $q = 3, 4$	exact
(B)	$r - 1 = (q - 1)n/2$: the boundary K -theory is a function of (q, n)	exact
(J)	witness families at $(q, n) = (2, 6), (2, 8), (2, 10)$: same boundary, different Jac	exact
(P)	Theorem 3: same K -theory, same type, coprime $ \text{Jac} $	exact
(S)	Table 2: the spread inside one boundary class	sampled
(R)	the Busemann cocycle is $q^\mathbb{Z}$ -valued (normalisation check only)	exact

Table 3: The verification battery (`boundary_blind.py`), 7/7 passing.

Row (R) deserves a caveat we state rather than bury: it verifies only that the cylinder measures $\nu(\Omega_v) = q^{1-d(O,v)}$ give a $q^\mathbb{Z}$ -valued Radon–Nikodym cocycle. That the *essential range* of that cocycle is attained — which is what Krieger’s theorem [21] needs — is quoted from [26, 27], not verified here; no finite computation could verify it.

6 Scope

Quoted, not proved here. Equations (1) and (2) and the rigidity observation [27]; the reconstruction theorem [9]; Krieger’s classification [21]; the Cuntz machine [10]; the forcing of III_1 in the continuum [12, 28, 4].

New here. Theorem 3 and Tables 1–2: the arithmetic layer, and the observation that K -theory and type together are still blind to it. The import of [27, 9] into p -adic holography, where no operator-algebraic invariant has previously been computed. The reading of $\lambda \neq 1$ in Section 4 as a diagnostic separating p -adic from continuum holography.

Open. (a) Whether the Cornelissen–Marcolli reconstruction [9], which recovers Y and hence $\text{Jac}(Y)$, can be made to recover the arithmetic *directly* — an explicit formula for $|\text{Jac}(Y)|$ from the boundary KMS data would be a genuine holographic dictionary entry for Frobenius traces. (b) The higher-rank case: for $\tilde{A}_n$ buildings the boundary type is $\text{III}_{1/q}$ or III_{1/q^2} according to the parity of n [11], and holographic constructions on buildings exist [13, 24], but none is arithmetic; the bulk invariant there is the rank-two critical group of [7, X], and we have not computed the corresponding witnesses. (c) Whether the periodicity of the modular flow has an entropic meaning.

Provenance

Unrefereed preprint. The three exact computations behind Theorem 3 — the Cuntz–Krieger K -groups, the bipartiteness, and the Smith normal form of the Hecke–Laplacian — are performed by the accompanying script and are reproducible in seconds. Everything else in Sections 1–4 is quoted, with the attribution set out in Section 6. We searched the physics literature for any prior use of an operator-algebraic invariant in p -adic AdS/CFT, and for any occurrence of type III_λ with $\lambda \neq 1$ in holography, and found none; a search cannot establish a negative, and we state this as the limit of our knowledge.

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